THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

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MATH3311/MATH5335

Mathematical Computing for Finance
Computational Methods for Finance

(1) TIME ALLOWED – 2 HOURS

(2) TOTAL NUMBER OF QUESTIONS – 4

(3) ANSWER ALL QUESTIONS

(4) THE QUESTIONS ARE OF EQUAL VALUE

(5) THIS PAPER MAY BE RETAINED BY THE CANDIDATE

(6) **ONLY** CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. i) Let $A \in \mathbb{R}^{n\times n}$.

a) Write down the LU factorization with pivoting of the matrix A and state the properties of the matrices.

b) Assume you want to solve the linear systems $A\mathbf{x}_k = \mathbf{b}_k$ for $k = 1, 2, \ldots, K$, where K is 'large'. Explain an efficient way of solving these linear systems.

c) What is the computational cost of solving these linear systems, given that computing the LU factorization costs $2n^3$ flops and computing a forward/backward substitution costs n^2 flops.

Correction: Computing LU costs (2/3)n^3 and not 2n^3

ii) Let $Q \in \mathbb{R}^{n\times n}$.

a) What property must Q satisfy to be an orthogonal matrix?

b) State the definition of the subordinate 2-norm of a matrix Q.

c) Show that if Q is orthogonal, then the 2-norm of Q is 1.

d) Show that if Q is orthogonal, then the 2-norm of the inverse matrix Q^{-1} is 1.

iii) Consider the matrix

$$D = B + \mathbf{u}\mathbf{v}^\top,$$

where

$$B = \begin{pmatrix} 2 & -1 & 0 & \ldots & 0 \\ -1 & 2 & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & -1 \\ 0 & \ldots & 0 & -1 & 2 \end{pmatrix}$$

and where $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are column vectors. Assume that n is large (say $n = 10^6$).

a) Why is explicitly forming D not a good idea?

b) Let $\mathbf{z} \in \mathbb{R}^n$ be given. Provide an algorithm which computes $D\mathbf{z}$ using the smallest possible number of additions and multiplications.

c) For a general $n \geq 1$, count how many additions and multiplications your algorithm requires to compute $D\mathbf{z}$.

2. i) Let $f \in C^2([a,b])$ and $x \in (a,b)$. Consider the expression

$$\frac{f(a+h)-f(a)}{h} = f'(a) + O(h). \tag{1}$$

a) Use Taylor series expansions to prove (1).

b) An analyst estimates the first derivative of a function using double precision in MATLAB at $a = 2 * 10^6$ using $h = 10^{-12}$ and gets the value 0. Is the analyst's approach correct?

ii) The standard normal cumulative distribution function (cdf) $\Phi : \mathbb{R} \to [0,1]$ is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \mathrm{e}^{-y^2/2}\, \mathrm{d}y. \tag{2}$$

The inverse of the standard normal cdf is given by $\Phi^{-1} : [0,1] \to \mathbb{R}$ such that $\Phi^{-1}(p) = x$, where $p = \Phi(x)$.

a) For a given $p \in [0,1]$, pose the problem of finding $\Phi^{-1}(p)$ as the solution of a nonlinear equation $f(x) = 0$.

b) Calculate the derivatives $f'(x)$ and $f''(x)$.

c) Prove that $f(x) = 0$ has a unique solution for any $0 < p < 1$.

d) If $\overline{x}$ is an approximate value of the root x, with error $|x - \overline{x}| \approx 10^{-6}$, estimate the error after one iteration of the Newton method, which is quadratically convergent.

iii) It is known that for any $0 \le n \ne m < N$ we have

$$\frac{2}{N} \sum_{k=0}^{N-1} \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right) \cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(m+\frac{1}{2}\right)\right)$$
$$= \begin{cases} 0 & \text{if } m \ne n, \\ 1 & \text{if } n = m. \end{cases}$$

(You do NOT have to proof these relations.)

In some applications the fast Fourier transform is replaced by the discrete cosine transform. One version of this transform is defined as follows: Let $x_0, x_1, \dots, x_{N-1} \in \mathbb{R}$ be unknown values and let

$$y_k = \sum_{n=0}^{N-1} x_n \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right) \quad \text{for } k = 0,1,\dots,N-1. \tag{3}$$

Assume you are given $y_0, y_1, \dots, y_{N-1}$. Find the inverse discrete cosine transform of (3) which allows you to compute $x_0, x_1, \dots, x_{N-1}$. Provide a proof for your answer.

3. i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F, so that

$$\mathbb{P}(Y \leq y) = F(y).$$

Let F^{-1} denote the inverse function of F. Let X be uniformly distributed in the interval $[0, 1]$. Show that the random variable $F^{-1}(X)$ has cumulative distribution function F.

ii) Transform the expected value

$$\mathbb{E}[h] = \int_{-\infty}^{\infty} h(x)\phi(x)\,\mathrm{d}x,$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$ is the standard normal density, into an integral over $[0, 1]$.

iii) An asset price S_t is assumed to follow a geometric Brownian motion

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma \mathrm{d}W_t, \quad t \in (0, T], \tag{4}$$

where W_t is a Wiener process, μ, σ are positive constants and S_0 is given.

a) What are the key properties of the Wiener process W_t?

b) Equation (4) can be discretized using

$$\frac{\Delta S_n}{S_n} = \mu\Delta t + \sigma\sqrt{\Delta t}\, Z_n,$$

where $Z_0, Z_1, \ldots, Z_{m-1}$ are independent, standard normal random variables, $t_n = n\Delta t$, $\Delta t = \frac{T}{m}$, and $\Delta S_n = S_{n+1} - S_n$.
Assume that you can generate independent samples $u_0, u_1, u_2, \ldots$ which are uniformly distributed in the interval $[0, 1]$. How can you use these samples to simulate the asset prices S_t at times t_n for $n = 0, 1, \ldots, m$?

c) Write a MATLAB code which creates a vector S of length $m + 1$ containing the value of S_n as its $n + 1$ element.

To answer this question, see the Matlab program stock1.m

4. The value $V(S,t)$ of an option on an underlying asset with price S at time t satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS\,\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2\,\frac{\partial^2 V}{\partial S^2} = rV \tag{5}$$

for $0 < S < \infty$ and $0 < t < T$. We assume that the prevailing risk-free interest rate r and the volatility σ are known positive constants.

Consider using (5) to price a **call** option, which gives the right to buy the underlying asset for the strike price K at expiry T.

i) Carefully explain what the "initial" conditions are for this problem.

ii) Carefully explain what the boundary conditions are for this problem.

iii) The change of variables to log-moneyness y and time to expiry τ is

$$y = \log(S/K) \qquad \text{and} \qquad \tau = T - t.$$

This gives $W(y,\tau) = V(S,t)$.

a) Find expressions for

$$\frac{\partial W}{\partial \tau}, \qquad \frac{\partial W}{\partial y}, \qquad \frac{\partial^2 W}{\partial y^2},$$

in terms of the partial derivatives of $V(S,t)$.

b) Show that the Black and Scholes PDE (5) is equivalent to

$$-\frac{\partial W}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\,\frac{\partial^2 W}{\partial y^2} = rW \tag{6}$$

in terms of the log-moneyness y and the time to expiry τ.

c) What are the initial conditions for the transformed problem?

d) What are the boundary conditions for the transformed problem?

iv) Let $y_{\max} > 0$ and consider the domain $y \in [-y_{\max}, y_{\max}]$ and $\tau \in [0,T]$. Set up a grid $(y_j, \tau_\ell) = (-y_{\max} + j\Delta y, \ell\Delta\tau)$ for $j = 0,1,2,\ldots,2(n+1)$ and $\ell = 0,1,2,\ldots,m$, where

$$\Delta y = \frac{y_{\max}}{n+1} \qquad \text{and} \qquad \Delta\tau = \frac{T}{m}.$$

Let $w_j^\ell \approx W(y_j,\tau_\ell)$ denote the approximate solution. At the time point $\tau_{\ell+1}$ and log-moneyness y_j give

a) central difference approximations of $O(\Delta y^2)$ for the partial derivatives

$$\frac{\partial W}{\partial y} \quad \text{and} \quad \frac{\partial^2 W}{\partial y^2}.$$

b) the backward difference approximation of $O(\Delta\tau)$ for the partial derivative $\frac{\partial W}{\partial \tau}$.

END OF EXAMINATION

Q1: i) a) $PA = LU$; $P, L, U \in \mathbb{R}^{n \times n}$

P .. permutation matrix

L .. unit lower triangular

U . upper triangular

b) $Ax_k = b_k$.

Compute LU factorization : $PA = LU$

Next we need to solve $Ax_k = b_k \iff PAx_k = LUx_k = Pb_k$.

To do so, first solve

$$Ly_k = Pb_k.$$

Then solve $Ux_k = y_k$.

c) Perform LU factorization once : $\frac{2}{3}n^3$ flops

Perform forward and backward substitution : $2n^2$ flops ← K times

Altogether : $\frac{2}{3}n^2 + 2Kn^2$ flops.

ii) a) $QQ^T = I$ identity matrix

$$\|Q\|_2 = \max_{\underline{x} \neq \underline{0}} \frac{\|Q\underline{x}\|_2}{\|\underline{x}\|_2} = \max_{\underline{x} : \|\underline{x}\|_2 = 1} \|Q\underline{x}\|_2$$

We have $\|Q\underline{x}\|_2^2 = (Q\underline{x})^T Q\underline{x} = \underline{x}^T Q^T Q \underline{x} = \underline{x}^T I \underline{x} = x^T x = \|\underline{x}\|_2^2$

Hence $\|Q\|_2 = \max_{\underline{x} \neq \underline{0}} \frac{\|Q\underline{x}\|_2}{\|\underline{x}\|_2} = \max_{\underline{x} \neq \underline{0}} \frac{\|\underline{x}\|_2}{\|\underline{x}\|_2} = 1$.

Similarly

$$\|Q^{-1}\underline{x}\|_2^2 = \|Q^T\underline{x}\|_2^2 = (Q^T\underline{x})^T Q^T\underline{x} = \underline{x}^T QQ^T\underline{x} = \underline{x}^T I \underline{x} = \underline{x}^T\underline{x} = \|\underline{x}\|$$

Thus

$$\|Q^{-1}\|_2 = \max_{\underline{x} \neq \underline{0}} \frac{\|Q^{-1}\underline{x}\|_2}{\|\underline{x}\|_2} = \max_{\underline{x} \neq \underline{0}} \frac{\|\underline{x}\|_2}{\|\underline{x}\|_2} = 1.$$

Q1:

iii) a) The matrix B is tridiagonal, wheras D is not sparse.

Storage for D is much larger than for B, u, v.

b)

$$D\underline{z} = B\underline{z} + \underline{u}\,\underline{v}^T\underline{z}$$

Compute: $\underline{v}^T\underline{z} = \alpha$ $\quad(-1+2n)$ flops

Compute: $\alpha\underline{u} = \underline{p}$ $\quad\approx$ n flops

Compute: $B\underline{z} = \underline{y}$

$y_k = -z_{k-1} + 2z_k - z_{k+1}$ $\quad$ 3 flops for $k = 2, \dots, n-1$

$y_1 = 2z_1 - z_2$ $\quad$ 2 flops

$y_n = 2z_n - z_{n-1}$ $\quad$ 2 flops

Add $\underline{p} + \underline{y}$ $\quad$ n flops

Altogether $2n-1 + n + 3(n-2) + 4 + n = 7n-3$ flops

Q2:

i) a) $f(x) = f(a) + (x-a) f'(a) + R(x)$

$$f(a+h) = f(a) + h f'(a) + R(a+h)$$

$$f'(a) = \frac{f(a+h)-f(a)}{h} + \underbrace{\frac{1}{h} \underbrace{R(a+h)}_{=O(h^2)}}_{O(h)} = \frac{f(a+h)-f(a)}{h} + O(h)$$

b) $a = 2 \cdot 10^6$

$h = 10^{-12}$

$$a+h = 2 \cdot 10^6 + 10^{-12} = 2 \cdot 10^6 \left[1 + \underbrace{\tfrac{1}{2} 10^{-18}}_{< \text{ machine epsilon}}\right]$$

$\Rightarrow 1 + \frac{1}{2} 10^{-18}$ is stored as 1 on computer using double precision

$\Rightarrow a+h$ is stored as $2 \cdot 10^{+6}$ on computer $\Rightarrow f(a+h) - f(a)$ yields 0 in double precision.

ANSWER: No.

ii) a) $\Phi^{-1}(p) = x \Leftrightarrow p = \Phi(x) \Leftrightarrow \Phi(x) - p = 0.$

Let $f(x) = \Phi(x) - p$. Then

$$\Phi^{-1}(p) = x \Leftrightarrow f(x) = 0.$$

b) $f'(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

$$f''(x) = -\frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Q2:

c) We have $f'(x) > 0 \quad \forall x. \Rightarrow$ strictly increasing

$$\lim_{x\to -\infty} f(x) = \lim_{x\to -\infty} \Phi(x) - p = -p < 0$$

$$\lim_{x\to \infty} f(x) = \lim_{x\to \infty} \Phi(x) - p = 1 - p > 0.$$

d) $err \approx 10^{-6}$

After one step : $\approx (10^{-6})^2 = 10^{-12}$.

iii) $$\frac{2}{N}\sum_{k=0}^{N-1} y_k \cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(m+\frac{1}{2}\right)\right)$$

$$= \frac{2}{N}\sum_{k=0}^{N-1}\sum_{n=0}^{N-1} x_n \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right)\cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(m+\frac{1}{2}\right)\right)$$

$$= \sum_{n=0}^{N-1} x_n \underbrace{\frac{2}{N}\sum_{k=0}^{N-1} \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right)\cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(m+\frac{1}{2}\right)\right)}_{=\begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } n = m \end{cases}}$$

$$= x_m.$$

Q3)

i) We have

$$F(a) = \mathbb{P}(X \leq F(a)) = \mathbb{P}(F^{-1}(X) \leq a).$$

ii) $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $\Phi(x) = \int_{-\infty}^{x} \phi(y)\,dy$

Set $z = \Phi(x)$. Then $\frac{dz}{dx} = \phi(x)$. Hence $x = \Phi^{-1}(z)$ and

$$\mathbb{E}(h) = \int_{-\infty}^{\infty} h(x) \underbrace{\phi(x)\,dx}_{=dz} = \int_{0}^{1} h(\Phi^{-1}(z))\,dz.$$

iii) a) $W(t)$ depends continuously on t for $0 \leq t \leq T$.

$W(0) = 0$ with probability 1

$W(t) - W(s) \sim N(0, t-s)$ for $0 \leq s < t \leq T$

$W(t) - W(s)$ and $W(v) - W(u)$ are independent for $0 \leq s < t < u < v \leq T$.

b) Set $z_n = \Phi^{-1}(u_n)$ for $n = 0, 1, 2, \ldots$ where Φ^{-1} is the inverse of

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}\,dy.$$

S_0 is given.

$$\frac{S_{n+1} - S_n}{S_n} = \mu \Delta t + \sigma \sqrt{\Delta t}\, z_n$$

$$\Rightarrow S_{n+1} = S_n\left[1 + \mu \Delta t + \sigma \sqrt{\Delta t}\, \Phi^{-1}(u_n)\right] \text{ for } n = 0, 1, 2, \ldots, m-$$

where S_n is an approximation of the price at time t_n.

Q4:

i) $V(S,T) = \max(S-K, 0) = \begin{cases} S-K & \text{if } S>K \\ 0 & \text{otherwise} \end{cases}$

strike price K

ii) $V(0,t) = 0, \quad t \in (0,T)$

$V(S,t) \to S - Ke^{-r(T-t)}$ as $S \to \infty$.

iii)

a) $$\frac{\partial W}{\partial \tau} = \frac{\partial V}{\partial t}\frac{\partial t}{\partial \tau} = -\frac{\partial V}{\partial t} \qquad (\tau = T-t)$$

$$\frac{\partial W}{\partial y} = \frac{\partial V}{\partial S}\frac{\partial S}{\partial y} = S\frac{\partial V}{\partial S} \qquad (S = Ke^{y})$$

$$\frac{\partial^2 W}{\partial y^2} = \frac{\partial}{\partial y}\left(S\frac{\partial V}{\partial S}\right) = S\frac{\partial V}{\partial S} + S\frac{\partial}{\partial y}\left(\frac{\partial V}{\partial S}\right)$$

$$= S\frac{\partial V}{\partial S} + S\frac{\partial^2 V}{\partial S^2}\frac{\partial S}{\partial y}$$

$$= S\frac{\partial V}{\partial S} + S^2\frac{\partial^2 V}{\partial S^2}$$

b) $$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2}{2}S^2\frac{\partial^2 V}{\partial S^2} = rV$$

$$-\frac{\partial W}{\partial \tau} + r\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\left(\frac{\partial^2 W}{\partial y^2} - \frac{\partial W}{\partial y}\right) = rW$$

$$-\frac{\partial W}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\frac{\partial^2 W}{\partial y^2} = rW$$

Q4:

iii) c)

$$W(y,0) = \max(Ke^y - K, 0)$$

d) $S \to 0^+ \iff y \to -\infty$

$S \to \infty \iff y \to \infty$

Hence

$$\lim_{y \to -\infty} W(y,\tau) = 0$$

$$W(y,\tau) = K\left(e^y - e^{-r\tau}\right) \text{ as } y \to +\infty.$$

iv)

a) $$\left.\frac{\partial W}{\partial y}\right|_{\substack{y_j \\ \tau_{l+1}}} = \frac{w_{j+1}^{l+1} - w_{j-1}^{l+1}}{2\Delta y} + O(\Delta y^2)$$

$$\left.\frac{\partial^2 W}{\partial y^2}\right|_{\substack{y_j \\ \tau_{l+1}}} = \frac{w_{j+1}^{l+1} - 2w_j^{l+1} + w_{j-1}^{l+1}}{\Delta y^2} + O(\Delta y^2)$$

b) $$\left.\frac{\partial W}{\partial \tau}\right|_{\substack{y_j \\ \tau_{l+1}}} = \frac{w_j^{l+1} - w_j^{l}}{\Delta \tau} + O(\Delta \tau)$$